

Options :

6406532335568. ✔ Parochialism

6406532335569. ✖ Ethnocentrism

6406532335570. ✖ Populism

6406532335571. ✖ Ottomanism

Question Number : 232 Question Id : 640653699432 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

[Organizational Culture] What term characterizes the attitude of the American team?

Options :

6406532335572. ✖ Parochialism

6406532335573. ✔ Ethnocentrism

6406532335574. ✖ Unipolarity

6406532335575. ✖ Egocentrism

Industry 4.0

Section Id :	64065349330
Section Number :	11
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	24
Number of Questions to be attempted :	24
Section Marks :	45

Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653103778
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 233 Question Id : 640653699442 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532335612.  YES

6406532335613.  NO

Sub-Section Number :	2
Sub-Section Id :	640653103779
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 234 Question Id : 640653699443 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is not a form of internet?

Options :

6406532335614. ✖ Industrial

6406532335615. ✖ Commercial

6406532335616. ✔ Philanthropic

6406532335617. ✖ Consumer

Question Number : 235 Question Id : 640653699445 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What is First Mile in a supply chain?

Options :

6406532335622. ✔ Movements from Vendor to Distribution centres

6406532335623. ✖ Movements from Distribution centres to stores

6406532335624. ✖ Movements from stores to customers

Question Number : 236 Question Id : 640653699446 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following facility layout is best suited for manufacturing several different products of similar types using the same production line?

Options :

6406532335625. ✖ Product layout

6406532335626. ✖ Process layout

6406532335627. ✖ Fixed position layout

6406532335628. ✔ Group layout

Question Number : 237 Question Id : 640653699449 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which type of layout is preferred for building construction?

Options :

6406532335631. ✖ Process layout

6406532335632. ✖ Product layout

6406532335633. ✔ Fixed product layout

6406532335634. ✖ None of these

Question Number : 238 Question Id : 640653699451 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which layout is apt for low volume production of highly customized products?

Options :

6406532335636. ✖ Product layout

6406532335637. ✔ Process layout

6406532335638. ✖ Fixed product layout

6406532335639. ✖ None of these

Question Number : 239 Question Id : 640653699452 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406532335640. ✖ IC engine

6406532335641. ✖ Electric car

6406532335642. ✔ Hyperloop

6406532335643. ✖ Steam locomotive

Sub-Section Number :	3
Sub-Section Id :	640653103780
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 240 Question Id : 640653699444 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

What are the inputs for an order-up-to-level calculation?

Options :

6406532335618. ✓ Sales forecast

6406532335619. ✓ Lead time

6406532335620. ✓ On-hand inventory

6406532335621. ✖ Safety stock

Question Number : 241 Question Id : 640653699454 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are considered in the Vehicle Routing Problem but not considered in TSP?

Options :

6406532335648. ✖ Subtour elimination

6406532335649. ✓ Any capacity restrictions

6406532335650. ✓ A specific starting point

6406532335651. ✖ Distance travelled restriction

Question Number : 242 Question Id : 640653699468 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following supply chain costs is/are affected by distribution network structure?

Options :

6406532335673. ✓ Inventory

6406532335674. ✓ Transportation

6406532335675. ✓ Facilities and handling

6406532335676. ✓ Information

6406532335677. ✖ Price

Sub-Section Number :	4
Sub-Section Id :	640653103781
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 243 Question Id : 640653699447 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Short Answer Question

Calculate the Metropolitan distance between two machines located at the coordinates A(5,10) and B (2,6) in a shop floor.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Range
Text Areas : PlainText

Possible Answers :

6.8 to 7.2

Sub-Section Number :	5
Sub-Section Id :	640653103782
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 244 Question Id : 640653699448 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Short Answer Question

Suppose a company wants to expand its logistics network and locate a new warehouse within a network of three existing facilities. The coordinates of the existing warehouses are: A(200,50), B(300,100), C(100,150) and the respective outbound goods volumes are 2500 units, 1300 units, 5000 units. Determine the location of the new warehouse (round off the answer to the nearest integer)

What is the x-coordinate of new warehouse?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

158

Question Number : 245 Question Id : 640653699450 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Short Answer Question

A factory works for two 8 hour shifts each day and produces a total of 24,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. What is the cycle time (in seconds) at any workstations (round your answer to one decimal place)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.3 to 2.5

Sub-Section Number :	6
Sub-Section Id :	640653103783
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 246 Question Id : 640653699453 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

- 6406532335644. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region
- 6406532335645. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity
- 6406532335646. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point
- 6406532335647. ✖ Optimal solution in a facility location problem using Euclidean method matches the centre of gravity in not of the cases

Question Number : 247 Question Id : 640653699465 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which are the two primary tasks of time series forecasting?

Options :

6406532335663. ✔ Identifying pattern

6406532335664. ✖ Identify variables of interest

6406532335665. ✔ Discovering relationships amongst variables of interest

Question Number : 248 Question Id : 640653699466 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following example(s) can be referred as seasonal pattern?

Options :

6406532335666. ✖ Economic recession

6406532335667. ✔ Increases Ceiling Fan sales during winters

6406532335668. ✔ Increased tourist footfall towards the end of the year

Sub-Section Number :	7
Sub-Section Id :	640653103784
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653699455 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (249 to 252)

Question Label : Comprehension

Bluescope company estimates its ordering cost at Rs 400 and carrying cost to be Rs. 2/year/unit. The estimated annual requirement is 10,000 units at a unit price of Rs. 10. Calculate the given subquestions:

Sub questions

Question Number : 249 Question Id : 640653699456 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the economic order quantity (EOQ)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 250 Question Id : 640653699457 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Calculate the number of orders per year.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 251 Question Id : 640653699458 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Calculate the annual inventory carrying cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 252 Question Id : 640653699459 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

While rechecked, Bluescope company finds that the order cost should have been 20% more than the currently estimated value. What should be the percentage increase in the EOQ in this case?

Enter your answer without the '%' symbol.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

9.4 to 9.6

Sub-Section Number : 8

Sub-Section Id : 640653103785

Question Shuffling Allowed : No

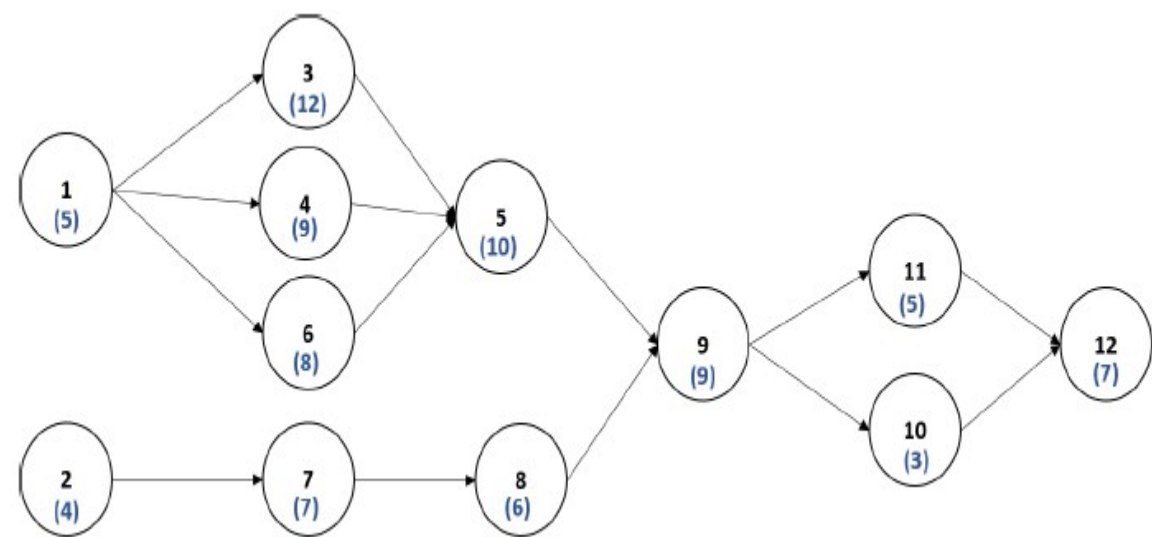
Is Section Default? : null

Question Id : 640653699460 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Question Numbers : (253 to 255)

Question Label : Comprehension

Consider the following precedence diagram. It has 12 activities. The numbers in blue and within the braces denote the respective activity times in minute.



Answer the given subquestions based on the above diagram.

Sub questions

Question Number : 253 Question Id : 640653699461 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Short Answer Question

What is the minimum cycle time of the system? (in minutes)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 254 Question Id : 640653699462 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Short Answer Question

What is the maximum cycle time of the system? (in minutes)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText

Possible Answers :

85

Question Number : 255 Question Id : 640653699463 Question Type : MCQ Is Question
Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction
Time : 0

Correct Marks : 1
Question Label : Multiple Choice Question

The factory works for one 8 hour shifts each day. Is it possible to achieve an output of 32 units/day,

Options :

6406532335658.  Yes

6406532335659.  No

Sub-Section Number :	9
Sub-Section Id :	640653103786
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 256 Question Id : 640653699464 Question Type : MCQ Is Question
Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Select the correct form of internet from column B corresponding to the example in Column A.

Column A	Column B
A1. RFID attached to inventory	B1. Consumer
A2. E-commerce	B2. Commercial
A3. Entertainment integrated device	B3. Industrial
	B4. Private

Options :

6406532335660. ✖ A1: B3, A2: B2, A3:B4

6406532335661. ✖ A1: B2, A2: B3, A3:B1

6406532335662. ✔ A1: B3, A2: B2, A3:B1

Question Number : 257 Question Id : 640653699467 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

What is the name of the forecasting method represented by the following set of equations? The notations represent the usual meaning.

$$F_{t+1} = a_t + b_t$$

Where

$$a_t = \alpha D_t + (1 - \alpha)(a_{t-1} - b_{t-1});$$

$$b_t = \beta(a_t - a_{t-1}) + (1 - \beta)b_{t-1}$$

Options :

6406532335669. ✖ Exponential smoothing

6406532335670. ✖ Winter's model for seasonality

6406532335671. ✔ Holt's method for trend

6406532335672. ✖ None of these

Question Number : 258 Question Id : 640653699469 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is not an example of digitization?

Options :

- 6406532335678. ✖ Creating the e -copy of a book using Google-Lens
- 6406532335679. ✔ Generating an automated daily sales report across the retail outlets in a state.
- 6406532335680. ✖ Converting a physical 2D sheet to CAD drawing
- 6406532335681. ✖ Converting physical invoice to e-copy

Sub-Section Number :	10
Sub-Section Id :	640653103787
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653699470 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (259 to 262)

Question Label : Comprehension

Ramu need to visit City-7. Currently he is stationed at City-1. He has a few alternative routes denoted by the arrows (arcs). There are other cities viz. 2, 3, 4, 5, 6 which falls on various routes between City-1 and City-7. The weights on the arcs represents the distances between respective nodes. However, there is no connection between City-2 and City-6. This scenario is represented in Fig-1.

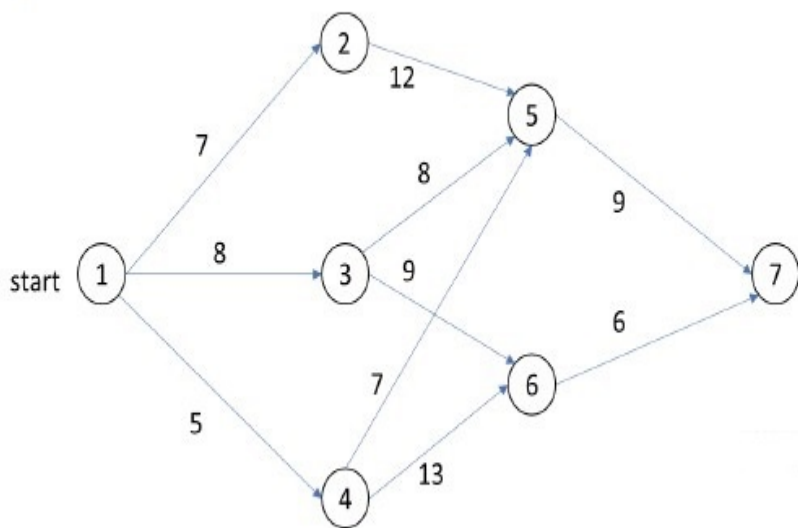


Fig-1

You need to apply dynamic programming (backward recursion) to identify the shortest route. Answer the given sub-questions:

Sub questions

Question Number : 259 Question Id : 640653699471 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What does the state variable represent?

Options :

6406532335682. ✖ The city Ramu should visit next in a stage

6406532335683. ✔ The beginning city in every stages

6406532335684. ✖ Distance covered upto a given stage

Question Number : 260 Question Id : 640653699472 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

What does the action set represents?

Options :

6406532335685. ✔ The city Ramu should visit next in a stage

6406532335686. ✖ The beginning city in every stages

6406532335687. ✖ Distance covered upto a given stage

Question Number : 261 Question Id : 640653699473 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the objective function value in Stage-2?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

Question Number : 262 Question Id : 640653699474 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Multiple Choice Question

What is the optimal route?

Options :

6406532335689. ✖ 1 -> 4 -> 6 -> 7

6406532335690. ✓ 1 -> 4 -> 5 -> 7

6406532335691. ✖ 1 -> 3 -> 6 -> 7

6406532335692. ✖ 1 -> 3 -> 5 -> 7

Sub-Section Number :	11
Sub-Section Id :	640653103788
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653699475 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (263 to 264)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There are total 100 seats in the aircraft. You have two classes of passengers-economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travellers follows a uniform distribution in [10,20]. Then answer the given subquestions:

Sub questions

Question Number : 263 Question Id : 640653699476 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

Question Number : 264 Question Id : 640653699477 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

83

Sub-Section Number : 12

Sub-Section Id : 640653103789

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653699478 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (265 to 267)

Question Label : Comprehension

Ramu puts up a stall for Diwali Light (Diya) every year. He builds a Diya at a cost of Rs. 2 and sells for Rs. 4. Any Diya not sold by Diwali will be discounted to 50 Paisa. From his experience, Ramu knows that the demand is Uniformly Distributed within [800, 1000] units. Then answer the given sub questions

Sub questions

Question Number : 265 Question Id : 640653699479 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the Critical Ratio? Round off your answer to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.5 to 0.6

Question Number : 266 Question Id : 640653699480 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the optimal order quantity that minimizes Ramu's total cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

912 to 916

Question Number : 267 Question Id : 640653699481 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question